

Conflict management for co-ordination between shift teams in Shanghai subway stations

Xin-an Zhang *Shanghai Jiao Tong University, China*

Huihua Liu *Shanghai Jiao Tong University, China*

Yushen Shi *Shanghai Jiao Tong University, China*

Dean Tjosvold *Lingnan University, Hong Kong*

Conflict among teams is inevitable in a multiteam system. Drawing on the theory of co-operation and competition, we proposed and tested a model linking the approaches teams adopt to conflict, inter-team co-ordination, and multiteam system performance. Data collected from 123 operation teams in 44 subway stations in Shanghai revealed that co-operative conflict management positively related to inter-team co-ordination, which in turn positively related to performance of the multiteam system. The relationship between competitive conflict management and inter-team co-ordination was insignificant and was moderated by co-operative conflict management. When co-operative conflict management was at a low level, competitive conflict management negatively related to inter-team co-ordination.

Keywords: competition, conflict management, cooperation, multiteam system, teams

Key points

- 1 This study finds co-operative conflict management between teams improves station performance through fostering inter-team co-ordination.
- 2 Competitive conflict management between teams is found not to be significantly related to station performance when co-operative conflict management was controlled for.
- 3 The relationship between competitive conflict management and inter-team co-ordination was insignificant and was moderated by co-operative conflict management. When co-operative conflict management was at a low level, competitive conflict management was negatively related to inter-team co-ordination.
- 4 Teams can be trained and oriented toward co-operative approaches to conflict to promote inter-team co-ordination and, subsequently, the performance of the multiteam system.

Correspondence: Xin-an Zhang, Antai College of Economics and Management, Shanghai Jiao Tong University, 1954 Huashan Road, Shanghai 200030, China; e-mail: xinanzhang@sjtu.edu.cn
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Introduction

Teams are thought to foster co-ordination of effort and resources across the functional boundaries needed to complete many tasks, improve internal efficiency, and adapt to environmental changes (Ashkenas 2002). Notwithstanding the prevalence of talk that teamwork has created the boundaryless organizations, studies suggest that team-based structures heighten the challenge of co-ordination on the inter-team level (e.g. Davison et al. 2012; Lanaj et al. 2013). Researchers have called for investigations on how teams in a multiteam system work together and, in particular, manage conflicts across team boundaries to improve the performance of the system (Luciano, DeChurch and Mathieu in press; Zaccaro, Marks and DeChurch 2012). Having teams compete against each other has long been thought to strengthen their motivation and productivity because they possess the incentives to win (Sherif and Sherif 1956). In this study, we argue that teams confront considerable conflict with each other that, if not well managed, frustrates the integration of their ideas and efforts. However, to the extent that teams can manage these conflicts co-operatively rather than competitively, the multi-team system as a whole can improve its performance through strengthened co-ordination. This study tests these hypotheses by examining the relationship between 123 operational shift teams in 44 subway stations in the city of Shanghai.

This study makes several important contributions to the literature. It examines the challenge of developing the conditions that foster inter-team co-ordination in a multiteam system and recognizes that how teams discuss and deal with their conflicts affects this co-ordination and then system performance (Davison et al. 2012). Drawing upon the theory of co-operation and competition (Deutsch 1973), we hypothesize that teams may adopt the co-operative, mutual benefit approach or the competitive, win-lose approach to managing their conflicts. Results challenge the traditional view that inter-group competition is useful for structuring the relationships between organizational teams. Findings suggest that these teams need information, resources, and support afforded by effective inter-team co-ordination to perform effectively. Whereas competitive conflict management among teams impedes inter-team co-ordination, co-operative conflict management among teams fosters inter-team co-ordination and attenuates the negative effect of competitive conflict management on inter-team co-ordination.

Conflict among teams

Teams have many reasons to encounter conflicts with other teams in a multiteam system (Hinsz and Betts 2012). For example in large new product projects, accounting teams may propose that the marketing team reduce its planned expenditures; marketing teams may be upset that the production team failed to deliver products to customers in a timely manner. Conflict also occurs between shift teams that are the sample for the present study. In a subway station, three different operation teams perform the same tasks in the same workplace where the daily job transfer from one shift to another provides abundant

opportunities for conflict. For example, shift teams develop opposing ideas about how to maintain the facilities used by all. Controversies may emerge regarding task allocation when one shift team leaves tough tasks to the next shift. Shift teams may try to persuade their managers that their shift is more effective and that they deserve more praise and support.

The theory of social identification provides an understanding for why conflicts between teams are so prevalent (Tajfel 1982). The theory suggests that individuals, when they are categorized as a member of a certain team, derive part of their self-concept from the membership of the team that they belong to, which leads to tendencies of ethnocentrism, in-group favoritism, and intergroup differentiation. These tendencies cause members of different teams to maintain an evaluative preference to their own groups and to emphasize the differences between their groups and other groups, making it difficult for them to take the perspective of each other, creating chances for conflicting attitudes and behaviors (Hogg and Terry 2000).

Conflict management among teams

Researchers in the West have joined those in the East in arguing that developing strong relationships is the key to understanding organizational dynamics as well as promoting organizational productivity (Gersick, Bartunek and Dutton 2000; Rousseau et al. 1998). In this study, we argue that teams in a system confront a great deal of conflict as teams try to complete their jobs, but this conflict can be managed in ways that help teams strengthen their co-ordination and system performance. Teams confront many task and relationship conflicts; both types of conflict can be handled effectively or ineffectively (Hinsz and Betts 2012). This study uses Deutsch's (1973) framework of co-operative and competitive approaches to conflict to understand how conflicts among teams may contribute to inter-team relations. In particular, we propose that the co-operative approach to conflict promotes inter-team co-ordination; the competitive approach to conflict impedes inter-team co-ordination; and the co-operative approach further attenuates the negative effect of the competitive approach on inter-team co-ordination. Then, we argue that inter-team co-ordination enhances performance of the multiteam system.

Conflict researchers have identified that it is how conflicts are managed that affects how constructive or destructive conflicts are. Conflict is not necessarily detrimental; it can be highly constructive, even essential to organizations depending on how it is approached and managed (Tjosvold 2008). However, managing conflict can itself be quite challenging. Researchers have long argued that even in the West organizational members are typically oriented toward avoiding difficult, emotionally charged issues that might question their underlying values and modes of working (Argyris and Schön 1996). Organizations are thought to force conflicts to be hidden, making conflict management difficult (Kolb and Bartunek 1992). This study uses the work of Deutsch (1973) to developing a clear definition of conflict among teams and an understanding on the major approaches teams have to discuss and deal with these conflicts.

Co-operative and competitive approaches to conflict management

Deutsch (1973) defined conflict as incompatible activities; inter-team conflict occurs when members of one team perceive their *actions* are interfered with, obstructed, or impeded by the *actions* of another team. This definition identifies how diverse teams can manage their conflicts for mutual benefit. Traditionally, conflict has been defined as opposing interests involving scarce resources and goal divergence and frustration. But this definition implies that conflict occurs between entities with incompatible goals where, if one benefits, the other entities cannot. Defining conflict as incompatible goals makes it more difficult to understand how conflicting parties might approach the conflict so that they both can move toward achieving their goals (Tjosvold 2008). Defining conflict as incompatible *activities* leaves open the possibility that conflicting parties may manage the conflict so that all parties benefit as they move toward achieving their goals.

Deutsch (1973) theorized that how conflicting parties believe their goals are related affects the dynamics and outcomes of conflict management and that co-operative goal interdependence is a foundation for constructive conflict. According to his theory of co-operation and competition, protagonists may conclude their goals are positively or negatively related. Co-operation is defined as positively related goals; it occurs when protagonists believe that as one of them moves toward goal attainment, the other protagonists are assisted in accomplishing their goals. In competition, protagonists believe their goals are negatively related so that if one succeeds the other cannot; they are in a 'win-lose' relationship.

Conclusions about how their goals are related to each other affects how protagonists interact during conflict and, thereby, the consequences of conflict management (Tjosvold 1998; Tjosvold, Wong and Chen 2014). With a co-operative approach to conflict, members of different teams are orientated to believing that the conflict problem needs common consideration and an integrative solution so that each team moves toward its goals. Although from different teams, members take a 'we are in it together' attitude toward conflict and seek solutions that will benefit all teams. With these positive expectations, members of different teams are more likely to discuss their opposing views directly and openly-mindedly where they both come to understand each other's opinions and integrate them into their own thinking and thereby reach agreement (Firth et al. 2015). Individuals from different teams develop psychological safety among themselves, which helps them feel free to express their own positions and doubts (Somech and Khalaili 2014); they use collaborative rather than contentious approaches to discuss and communicate varying perspectives because they believe they are confronted with a common task (Luciano, DeChurch and Mathieu in press). These interaction dynamics induced by co-operative goals are labeled the co-operative approach to conflict management.

Members of different teams may also believe that their goals are competitive, concluding that one team's successful goal attainment makes other teams less likely to reach their own goals (Tjosvold, Wong and Chen 2014). To the extent that different teams assume competitive goals, they treat conflict as a win-lose contest and engage in such actions as overstating their own position to achieve their goals and demanding that other teams

agree with their position (Firth et al. 2015). Members from different teams may be able to develop an accurate understanding of each other's positions, but they are very reluctant to incorporate other views into their own (Cuijpers, Uitdewilligen and Guenter 2015; Marks et al. 2005). Because of expectations that other teams will not reciprocate openness and concessions and may even obstruct their efforts, members from different teams are often inflexible, resulting in deadlocks or the imposing of a solution by the more powerful protagonist (de Vries et al. 2016). They are locked in a battle over who will win and who will lose. The interaction process induced by competitive goals is called the competitive approach to conflict.

Typically, teams in conflict can and often possess both co-operative and competitive goals (Sherif 1961). As such, they can approach conflict both co-operatively and competitively (Zhang, Cao and Tjosvold 2011). Though typically negatively correlated, these approaches are considered alternatives rather than opposite ends of one continuum. To the extent that teams focus on their co-operative goals and approach conflict co-operatively, they are more able to co-ordinate their ideas and efforts.

Conflict management and inter-team co-ordination

A multiteam system needs to co-ordinate and integrate efforts of diverse teams to perform effectively (Ambrosini, Bowman and Burton-Taylor 2007; Davison et al. 2012; Hoegl, Weinkauff and Gemuenden 2004). Inter-team co-ordination refers to the extent to which ideas and efforts from members of different interdependent teams can be integrated (Davison et al. 2012). This need to integrate ideas and efforts from diverse teams presents a great deal of challenges for managing conflicts among teams.

Conflicts themselves do not mean that teams cannot co-ordinate effectively. The co-operative approach is a medium through which teams communicate open-mindedly and are willing to consider each other's opinions (Firth et al. 2015). This open communication contributes to more accurate understanding of each other's ideas and more awareness of each other's contribution. Consequently, teams reduce their tensions, believe that they can trust each other, and feel reconciled and supported (Somech and Khalaili 2014). The strengthening of their relationships in turn eases the process of co-ordinating and combining diverse efforts and ideas from different teams for effective solutions. Moreover, managing conflict co-operatively is a foundation to help teams reflect upon and learn from other teams' experiences, so that more effective solutions can be developed (Hoegl, Weinkauff and Gemuenden 2004).

In contrast, competitive conflict management among teams impedes inter-team co-ordination. When members of different teams view each other's goals as competitive, they are more aware that they belong to different teams and have a different identity (Sherif 1961). As a result of the increased salience of individual team identity, members of each team feel more within-group favoritism and out-group hostility (Hogg and Terry 2000). Different teams then tend to try to impose their own opinions on each other. Teams argue for the rightness of their own ideas and the weaknesses of other teams (Cuijpers, Uitdewilligen and Guenter 2015); they believe their ideas are superior and dismiss potentially

valuable input from members of other teams. Members of an individual team are not fully aware of the distribution of expertise among members of all teams, and that further disrupts co-ordination and integration (Hoegl and Weinkauff 2005). Moreover, competitive interactions undermine feelings of trust and confidence that different teams can develop solutions together, further impede integrating ideas and synchronizing effort and resources (de Vries et al. 2016).

As the competitive approach impedes inter-team co-ordination, this negative relationship is likely to be mitigated when teams simultaneously adopt the co-operative approach to manage conflict among them. As suggested above, the competitive approach is disruptive because it augments the out-group hostility resulted from social categorization process. However, when teams also adopt co-operative conflict management, the out-group hostility is likely to be reduced, because teams in co-operation can realize that they belong to an even larger system in which all teams share a common goal (Sherif 1961). Past research has indicated that a common goal helps teams reduce tensions, align their behavior, and co-ordinate (Davison et al. 2012; Hoegl, Weinkauff and Gemuenden 2004; Marks et al. 2005).

In a similar vein, the theory of optimal distinctiveness theory suggests individuals are most committed to a group that offers them the identity that can successfully satisfy their needs for both affiliation and distinction (Brewer 1991). Adopting both competitive and co-operative approaches can help team members reach such an optimal identity. Whereas competition among teams increases within-team cohesion and make team identity salient to satisfy team members' need for distinction, co-operation among teams increases the salience of a larger system of which all teams are interdependent components, which satisfies team members' need for affiliation. When members of different teams enjoy the feeling of optimal distinctiveness, they are more likely to exhibit behaviors that contribute to their own teams and to the system as a whole, and inter-team co-ordination is consequently improved.

Based on the above research and reasoning, we hypothesize:

Hypothesis 1a: Co-operative conflict management between teams is positively related to inter-team co-ordination.

Hypothesis 1b: Competitive conflict management between teams is negatively related to inter-team co-ordination.

Hypothesis 1c: The negative relationship between competitive conflict management and inter-team co-ordination is moderated by co-operative conflict management such that the negative relationship becomes weaker for teams that also adopt high levels of co-operative conflict management.

Inter-team co-ordination for system performance

We further argue that the system as a whole becomes more effective when individual teams are highly co-ordinated. Although different teams in a system share a common goal, they may pursue proximal goals because they are distinguishable entities (Arrow

and McGrath 1995). Inter-team co-ordination heightens the salience of the common goal for all teams, which helps them unify their resources and efforts toward the super-ordinate goal (Davison et al. 2012). In addition, inter-team co-ordination helps break-down team boundaries and integrate wisdom and efforts of diverse members from different teams (Ancona and Caldwell 1992; Marrone 2010). Because members from different teams are often more heterogeneous than those in a single team, the integration of more diverse inputs from different teams is likely to create a better solution that facilitates the operation of the system (Milliken and Martins 1996). Moreover, inter-team co-ordination leaves the space for teams to jointly determine how to reach the common goal, which engenders in them a sense of participation in decision-making. According to self-determination theory (Ryan and Deci 2000), when teams feel that they are performing tasks in a way that is determined by themselves, they become more committed to the goal and more persistent in front of difficulties (Gagné and Deci 2005). Taken together, we hypothesize a positive relationship between inter-team co-ordination and system performance.

Hypothesis 2: Inter-team co-ordination is positively related to system performance.

Conflict management in China

The empirical basis supporting the differential effect of co-operative vs competitive approaches to conflict management has been largely developed in North America. The utility of conflict, as well as the theories to analyze conflict, cannot be assumed to apply to a collectivist society like China (Hofstede 1993). Despite conflict not generally being valued in China, recent research suggests that Chinese people can discuss conflict productively. Leung, Koch and Lu (2002) has argued that, although Chinese people may use harmony-seeking behavior as a way to avoid potential problems, harmony also represents a genuine concern for feelings of intimacy, trust, compatibility, and mutually beneficial behaviors. With this motive, Chinese people discuss their conflicts openly to strengthen their relationships. This study tests the extent to which the Deutsch's western model of conflict management can be usefully applied to inter-team level in a Chinese context.

Method

Participants and procedures

Data were collected from 50 subway stations in the city of Shanghai, which has seen enormous changes in public transport in the last two decades. Each station was run by three shift teams on a daily basis, and each team has a leader. The average team size was 4.7 (SD = 1.3). The responsibilities of a shift team included daily operations, facility maintenance, and customer service. Before each team went off its shift, a meeting must be held with staff of the next shift team to hand over tasks and facilities. The meeting was important because it was part of the regulations to ensure operational safety and service quality.

Questionnaires were completed in two separate meeting rooms for team leaders and members, respectively, at the headquarters of the subway company. All team leaders and members of these 50 stations were invited to visit the headquarters to finish the survey during two assigned days when they went off their shifts. One another half day was arranged for those who did not attend the survey one week later. Team members rated items of inter-team conflict management and co-ordination; team leaders rated performance of the station as a multiteam system. Because the level of analysis was at the station level, we retained a station in the sample only when it had responses from no fewer than two team leaders as well as no fewer than two shift teams. Further, we required that each retained shift team had responses from no fewer than three members. With these criteria, the final sample consisted of 44 stations with 120 team leaders (mean = 2.7, SD = 0.8), 123 teams (mean = 2.8, SD = 0.8) and 492 members (mean = 11.2, SD = 4.1). The response rates were 85.1%, 90.9%, and 93.2% for team members, team leaders, and shift teams, respectively. No significant differences were identified between teams kept in the sample and those that dropped off in terms of main team demographics.

Forty-nine percent of the team members were male; the average age was 28.5 years old (SD = 9.4); 4% had less than a high school qualification, 38% had a high school qualification, 54% had a junior college or professional school degree, 4% had a university or higher level degree. The average organizational tenure was 3.5 years (SD = 4.2), and the average tenure on their team was 1.9 years (SD = 2.3).

Measures

Inter-team conflict management

We modified the scales previously developed to measure within-team conflict management to assess inter-team conflict management (Alper, Tjosvold and Law 1998). These scales were developed from a series of experimental and field studies and we adjusted item referent from 'team members' to 'members from different shifts' to accommodate our purpose. Participants were first asked to think about how people from different shifts interacted with each other when they negotiated their differences. Then they were asked to rate on a 7-point scale (1 = 'strongly disagree', 7 = 'strongly agree') indicating their degree of agreement with the statement in scale items. The co-operative conflict management scale consisted of five items. See the Appendix for a listing of all items. The reliability of the scale was of 0.87 at the individual level and 0.91 at the station level, and checks for aggregation yielded acceptable values ($r_{wg} = 0.95$; ICC1 = 0.12; ICC2 = 0.58; $F(43, 448) = 1.72, p < 0.01$).

The competitive conflict management scale consisted of four items. See Appendix for a listing of all items. The reliability of the scale was of 0.85 at the individual level and 0.88 at the station level. Checks for aggregation were acceptable ($r_{wg} = 0.94$; ICC1 = 0.10; ICC2 = 0.52; $F(43, 448) = 1.66, p < 0.01$). Results of confirmatory factor analysis using individual level data indicated discriminant validity for the two approaches to conflict

management ($\chi^2 = 117.36$, $df = 26$; RMSEA = 0.06; CFI = 0.98; NNFI = 0.97). We noted here that the relatively large Chi-square value was probably due to the size of the sample ($n = 492$) used (Bentler and Bonett 1980), and combining the two factors into a single dimension resulted in significant increase in Chi-square value ($\Delta\chi^2 = 778.65$, $p < 0.01$).

Inter-team co-ordination

We assessed inter-team co-ordination using five items. These items are based on the scale of inter-team co-ordination used by Hoegl, Weinkauff and Gemuenden (2004) and we revised their items to fit the research situation. See Appendix for a listing of all items. Shift members rated the degree of their agreement to these items on a response scale ranging from 1 = 'strongly disagree', to 7 = 'strongly agree'. The reliability of the scale was of 0.86 at the individual level and 0.92 at the station level, and checks for aggregation yielded acceptable values ($r_{wg} = 0.96$; ICC1 = 0.11; ICC2 = 0.56; $F(43, 430) = 2.09$, $p < 0.01$).

System performance

We modified the team performance scale previously used by Lu, Tjosvold and Shi (2010) to assess the performance of a subway station as a multiteam system. We changed the item referent to fit the current research. See Appendix for a listing of all items. Team leaders responded to these items on a 7-point scale (1 = 'strongly disagree', and 7 = 'strongly agree'). The reliability of the scale was of 0.88 at the individual level and 0.91 at the unit level, and checks for aggregation yielded acceptable values ($r_{wg} = 0.91$; ICC1 = 0.13; ICC2 = 0.27; $F(43, 79) = 1.59$, $p < 0.05$).

Results

We used structural equation modeling to test our hypotheses. According to the two-step procedure recommended by Anderson and Gerbing (1988), the measurement model should be examined prior to the simultaneous estimation of the measurement and structural models. Therefore, we conducted a series of confirmatory factor analyses to examine whether inter-team co-ordination, co-operative conflict management, competitive conflict management, and station performance would load on four distinct factors on the station level of analysis.

Confirmatory factor analysis

We conducted confirmatory factor analysis using LISREL 8.80 (Jöreskog and Sörbom 1993). The baseline 4-factor model was tested against alternative 3-factor structures in which two factors were combined into a single dimension. As shown in Table 1, the proposed 4-factor model provided the best fit to the data compared with the other models investigated. Specifically, the χ^2/df ratio (1.25) was below 2; both CFI (0.98) and NNFI (0.97) were not < 0.95 ; SRMR (0.068) was < 0.08 , all met the cutoff rules suggested by Hu and Bentler (1999). In contrast, each other of the alternative models improved the

Table 1 Results of CFA model comparison

Models	df	Model χ^2	$\Delta\chi^2$	CFI	NNFI	SRMR
1 Baseline 4-factor model	146	183.04	—	.98	.97	.068
2 Null model	165	967.06	784.02**	.48	.01	.770
3 Co-operative and competitive conflict management combined	149	224.25	41.21**	.96	.95	.077
4 Co-operative conflict management and inter-team co-ordination combined	149	367.12	184.08**	.87	.85	.140
5 Competitive conflict management and inter-team co-ordination combined	149	333.25	150.21**	.89	.88	.200
6 Co-operative conflict management and station performance combined	149	237.77	54.73**	.95	.94	.083
7 Competitive conflict management and station performance combined	149	239.97	56.93**	.95	.94	.086
8 Inter-team co-ordination and station performance combined	149	329.29	146.25**	.89	.88	.130

$N = 44$. ** $p < 0.01$. $\Delta\chi^2$ is the change of χ^2 compared with the baseline model.

Table 2 Means, standard deviations, coefficient alphas, and inter-correlations between study variables

	Mean	SD	1	2	3	4
1 Station effectiveness	5.43	.55	(.91)			
2 Inter-team co-ordination	5.70	.41	.49**	(.92)		
3 Co-operative conflict management	5.68	.35	.31*	.74**	(.91)	
4 Competitive conflict management	2.56	.46	-.31*	-.66**	-.67**	(.88)

$N = 44$. * $p < 0.05$; ** $p < 0.01$. Estimated reliabilities are in parentheses along the main diagonal.

Chi-square value significantly (for all alternative models, the significance level of the Chi-square change was $p < 0.01$). These results together suggested that the baseline measurement model provided the best approximation of the data, and thus we concluded that the four constructs were captured by the 19 items and could be distinguished reasonably well on the station level of analysis.

Hypothesis tests

Table 2 presents means, standard deviations, and correlations among study variables. As shown, both co-operative conflict management ($r = 0.74$, $p < 0.01$) and competitive conflict management ($r = -0.66$, $p < 0.01$) correlated significantly with inter-team co-ordination, providing initial support for hypotheses 1a and 1b. Further, hypothesis 2 was supported by the significant correlation between inter-team co-ordination and station

performance ($r = 0.49, p < 0.01$). Below we report the results of structural equation modeling for more parsimonious tests of hypothesis 1a, hypothesis 1b, and hypothesis 2, following the suggestions of James, Mulaik and Brett (2006).

The hypothesized baseline model represents a full mediation model in which the effects of conflict management on unit effectiveness are expected to occur through inter-team co-ordination. The baseline model was tested against three alternative models in which the baseline model was nested. Specifically, model 2 included a direct path from co-operative conflict management to station performance; model 3 included a direct path from competitive conflict management to station performance; and model 4 included both direct paths.

The hypothesized baseline model provided a good fit to the data (Table 3). All fit indices including chi-square (183.71), df (148), Chi-square/ df ratio (1.24), CFI (0.98), NNFI (0.98), and RMSEA (0.07) met traditional cutoff rules (Hu and Bentler 1999). In comparison with other three alternative models, the baseline model did not significantly increase Chi-square. Because the baseline model was nested in other three models, and according to the principle of parsimony, we finally selected the baseline model to test our hypotheses.

Figure 1 shows the path coefficients for the baseline model. Although the significant path from co-operative conflict management to inter-team co-ordination ($\gamma = 0.65, p < 0.01$) supported hypothesis 1a, the path from competitive conflict management to

Table 3 Results of SEM model comparison

Models	df	Model χ^2	$\Delta\chi^2$	CFI	NNFI	SRMR
1 Co-operative conflict + Competitive conflict → Inter-team co-ordination → Station performance(baseline model)	148	183.71	—	.98	.98	.071
2 Co-operative conflict + Competitive conflict → Inter-team co-ordination → Station performance; Co-operative conflict → Station performance	147	183.03	-.68	.98	.98	.068
3 Co-operative conflict + Competitive conflict → Inter-team co-ordination → Station performance; Competitive conflict → Station performance	147	183.52	-.19	.98	.98	.070
4 Co-operative conflict + Competitive conflict → Inter-team co-ordination → Station performance; Co-operative conflict + Competitive conflict → Station performance	146	183.04	-.67	.98	.97	.068

$N = 44$. * $p < 0.05$. $\Delta\chi^2$ is the change of χ^2 compared with the baseline model.

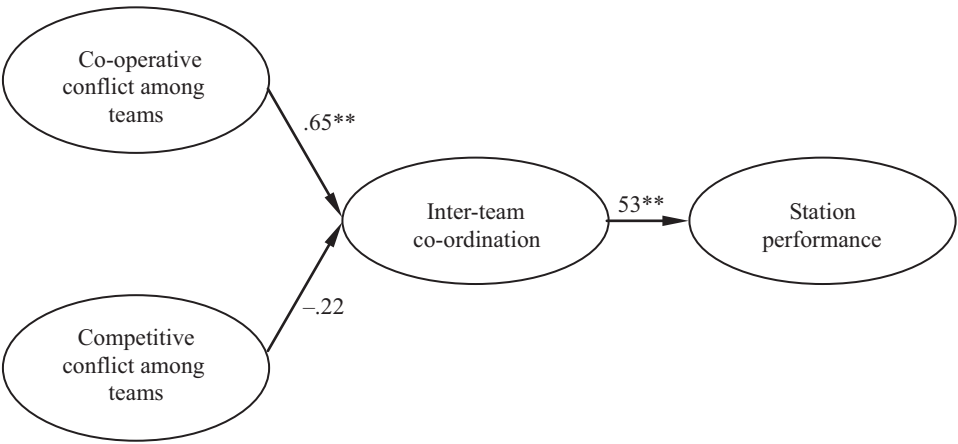


Figure 1 Path coefficients for hypothesized model
** $p < 0.01$.

inter-team co-ordination was non-significant ($\gamma = -0.20$, ns), leaving hypothesis 1b unsupported. The path from inter-team co-ordination to unit effectiveness was significant ($\beta = 0.53$, $p < 0.01$), which supported hypothesis 2. Because including a direct path from co-operative conflict management to station performance did not improve the model fit (M2), inter-team co-ordination was suggested to fully mediate the effect of co-operative conflict management on station performance. We further tested whether this indirect effect was significant using Sobel's test (Sobel 1982), results of which ($z = 2.23$, $p < 0.05$) suggested the indirect effect from the co-operative conflict management to station performance through inter-team co-ordination was significant.

Finally, to test hypothesis 1c regarding the joint effects of the co-operative and competitive approaches, we followed Ping's (1996) latent product procedure because it recovers parameter values well using LISREL (Cortina, Chen and Dunlap 2001). Specifically, we computed the product of the sums of respective indicators for the co-operative and competitive approaches, and used the product as the sole indicator for the latent product. We centralized the sum value for each conflict management approach before computation of the product, and fixed correlations between each approach and their product (i.e. PH) at zero as a result of centering. The new model with the interactive product fitted the data quite well: Chi-square (180.32), df (164), Chi-square/ df ratio (1.10), CFI (0.99), NNFI (0.99), and RMSEA (0.05). The coefficient for the latent product was 0.19 ($p < 0.05$), suggesting hypothesis 1c was supported. To illustrate the nature of the moderation, we computed simple slopes for high (+1SD) and low (-1SD) levels of the co-operative approach following the procedure recommended by Aiken and West (1991). As is shown in Figure 2, the relationship between the competitive approach and inter-team co-ordination was significant

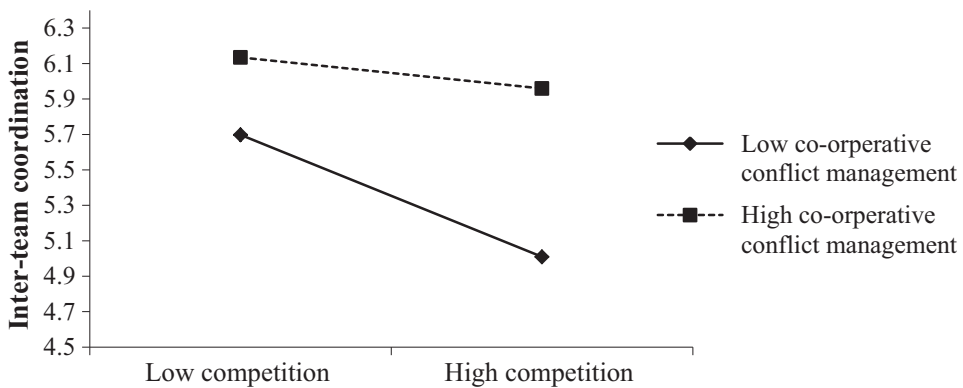


Figure 2 Joint effects of co-operative and competitive conflict management

when the co-operative approach was low ($\gamma = -0.40, p < 0.01$), and became non-significant when the co-operative approach was high ($\gamma = -0.02, ns$).

Discussion

This study complements considerable recent research on teams in organizations by investigating how teams can work together effectively when they are performing interdependent tasks and how their co-ordination affects unit performance. It addressed this issue from the perspective of conflict management. Based on data collected from shift teams in Shanghai subway stations, findings confirm that how teams approach their conflicts does influence the interactions between them and the performance of all teams as a system. Specifically, co-operative conflict management between teams improves station performance through fostering inter-team co-ordination, whereas competitive conflict management between teams was found not to be significantly related to inter-team co-ordination when co-operative conflict management was controlled for. In addition, moderation analysis found the effect of competitive conflict management was contingent on co-operative conflict management. That is, the negative relationship between competitive conflict management and inter-team co-ordination was significant only when co-operative management was at a low level.

Organizations must find ways to co-ordinate and integrate diversified efforts from teams when they rely on multiteams to get tasks finished (Davison et al. 2012; Marrone 2010). Our results suggest that by managing conflicts between teams co-operatively, teams can turn their differences and controversies into co-ordinated actions and, thus, improve the effectiveness of the whole system. Team co-operative conflict management also mitigates the potential negative effect of between-team competition. These findings join the stream of research responding to the call made by Zaccaro, Marks and DeChurch (2012) to examine between-team dynamics and help answer the question regarding *how* teams can work together effectively as a system.

Findings also have important implications for the theory of conflict management. Prior studies have documented the effect of intra-team conflict management on team effectiveness (Tjosvold 2008; Tjosvold, Wong and Chen 2014); however, seldom have organizational scholars investigated the effect of between-team conflict management. Because conflict is ubiquitous between teams when teams work interdependently, our findings extend conflict management theory from intra-team to inter-team level of analysis. Consistent with prior findings on team level, this research confirms that inter-team co-operative conflict management improves the performance of all teams as a system. Moreover, we identified inter-team co-ordination as an intervening mechanism through which conflict management influences unit effectiveness, thus, helping clarify how conflict management works across different teams.

The competitive approach to conflict among teams was hypothesized to negatively relate to inter-team co-ordination and then system performance. Although supported by correlation results, this study's structural equation modeling results suggested that the competitive approach does not make a significant contribution to inter-team co-ordination when the co-operative approach to conflict is controlled. One possible explanation is that the non-significant correlation might be due to the intra-team motivation effect generated from inter-team competition (Cuijpers, Uitdewilligen and Guenter 2015). As suggested by social identity theory, members of a team become more cohesive and committed to the team when they feel threatened by competitors (Hogg and Terry 2000). Competition among shift teams may have increased this motivation which led to better performance for each individual team, which then counterbalanced the expected negative effect of competitive conflict management among teams and made the relationship unobservable.

Findings also speak to the theory of optimal distinctiveness. In the context of a multi-team system, the competitive approach to between-team conflict is likely to augment the out-group hostility resulting from the social categorization process. This study demonstrated that by co-operatively managing between-team conflict, members of individual teams could develop an optimal level of team identity which simultaneously satisfied their needs for both affiliation and differentiation, and teams consequently co-ordinated effectively. Because prior studies on optimal distinctiveness theory primarily relied on experiments in laboratory settings (e.g. Hornsey and Hogg 1999; Leonardelli and Loyd 2016; Pickett and Brewer 2001), this study contributes to the theory by lending evidence drawn from a field study specifically in the context of a multiteam system that extends the external validity of the theory.

In addition to developing theoretical understanding, the hypotheses, if they can continue to be supported, have important practical implications for human resource management in multiteam systems. Results documented that the co-operative approach to conflict can be very useful for developing inter-team co-ordination and system performance. One possible avenue to substantiate these findings is through training (Delaney and Huselid 1996). Teams can be trained and oriented toward co-operative approaches to conflict. Employees from different teams can together develop shared goals, integrate roles

and common tasks, and share reward distributions that build commitment to co-operative goals (de Vries et al. 2014). In addition, employees can be trained in the skills of co-operative discussion across teams. With this training, employees from different teams become willing to express their ideas and positions, ask each other for more information and arguments, and try to combine the best ideas to create the highest quality solution. Training teams of skills of co-operative conflict management is of particular importance when competition between teams is prevalent or inevitable. Teams are often motivated or encouraged to compete against each other as a means to achieving better performance or more reward. Consequently, each team can easily adopt competitive approach to conflicts with other teams. Our findings indicate that this competitive approach to between-team conflict could be detrimental for the performance of all teams together as a system when co-operative conflict management is absent. Training teams of co-operative conflict management skills has the potential to mitigate this negative effect so that between-team competition will not harm the performance of the whole system.

Another avenue for multiteam organizations to build advantage from our findings is through promoting co-operative goal interdependence among teams. Past research indicated that how parties in a conflict perceive their goals are related influences what approach they adopt to managing the conflict (Deutsch 1973). When different teams perceive that they have co-operative goals, they believe the success of other teams will benefit themselves and are more likely to take co-operative approach to managing conflicts with other teams. We suggest that organizations set up and communicate a superordinate goal to all of the teams, or to update their reward system to promote co-operative goal interdependence among teams.

Limitations

The results of this study are, of course, limited by the sample and operations. The data are self-reported and subject to biases, and may not accurately describe the relationships, although prior research suggests that self-reported data are not as limited as commonly expected (Spector 1994). These data are also correlational and do not provide direct evidence of causal links between conflict approaches, inter-team co-ordination, and system performance. Objective indicators such as passenger satisfaction and facility failure rate would improve the measure of unit's performance. However, the data, as they were collected from employees and supervisors, should reduce the possibilities of common method error. Developing different sources for the independent and dependent measures should reduce the possibilities of same source method as an alternative explanation of the results.

The sample size was relatively small and drawn from a single company; we encourage future research to replicate this study with more teams from diversified companies and industries to generalize our findings. Finally, we did not examine boundary conditions for the effect of inter-team conflict management, which might be augmented or weakened under certain circumstances. For example, one condition, task interdependence among teams, may moderate the relationship such that the effect of between-team conflict

management on system performance becomes even stronger for highly interdependent teams than for lowly interdependent teams.

In summary, this study found that appropriate conflict management between teams is a source of effectiveness when units rely on multiteams to work together for a common goal. Interdependent work teams are suggested to improve their co-ordination and subsequently overall performance by encouraging co-operative conflict management between them.

Xin-an Zhang (PhD, Shanghai Jiao Tong Univ) is currently a professor of management in Antai College of Economics and Management, Shanghai Jiao Tong University.

Huihua Liu (MA, Zhengzhou Univ) is currently a doctoral student in Antai College of Economics and Management, Shanghai Jiao Tong University.

Yushen Shi (BA, Northeast Univ, China) is currently a doctoral student in Antai College of Economics and Management, Shanghai Jiao Tong University.

Dean Tjosvold (PhD, Minnesota) is an emeritus professor of management in the Department of Management, Lingnan University, Hong Kong. He has published over 200 articles and 20 books on co-operation and competition, managing conflict, leadership and power.

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Appendix 1

Scales used in the study

Co-operative conflict management (adapted from Alper, Tjosvold and Law 1998)

- 1 Members from different shifts encourage a ‘we are in it together’ attitude;
- 2 Members from different shifts seek a solution that will be good for all of us;
- 3 Members from different shifts treat conflict as a mutual problem to solve;
- 4 Members from different shifts work so that to the extent possible we all get what we really want;
- 5 Members from different shifts combine the best of positions to make an effective decision.

Competitive conflict management (adapted from Alper, Tjosvold and Law 1998)

- 1 Members from one shift demands that the people from other shifts agree to their position;
- 2 Members from different shifts want other teams to make concessions but do not want to make concessions themselves;
- 3 Members from different shifts treat conflict as a win-lose contest;
- 4 Members from different shifts overstate their positions to get its way.

Inter-team co-ordination (adapted from Hoegl, Weinkauff and Gemuenden 2004)

- 1 Generally speaking, employees from the various shifts are very satisfied with their work with each other.
- 2 Shift members feel a strong commitment to their joint work for the station.
- 3 Members of different shifts feel highly committed to the goals of their station.
- 4 The way the various shifts manage their work with each other inspires all the station employees to better job performance.
- 5 All things considered, the shifts are highly pleased with the way they manage their work with each other.

Station performance (adapted from Lu, Tjosvold and Shi 2010)

- 1 Together, the station has a high work performance.
- 2 Most of the joint tasks among shifts are accomplished quickly and efficiently.
- 3 The station shift teams always set a high standard of task accomplishment.
- 4 Shift teams together always achieve a high standard of task accomplishment.
- 5 Shift teams together almost always beat their targets.